Cover Letter

Dear Dr. Ahsan, Guest Editor of Systems,

We would like to submit the enclosed manuscript entitled “Early-Career Choices in the Era of Generative AI: Human Capital Depreciation, Career Anxiety, and Occupational Adaptation Among Fresh Graduates” for consideration in Systems, for the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” (Guest Editor: Dr. Ali Ahsan). The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Generative artificial intelligence is most capable at precisely the tasks from which early careers have always been built, and the students now leaving university are the first cohort to choose occupations under that condition. The demand side of this collision is being measured intensively; the supply side — what the entrants themselves perceive, feel and then do — is nearly unstudied, even though early-career scarring is among the most persistent phenomena in labor economics. This manuscript treats the fresh graduate as an adaptive agent in a socio-technical labor system: perceived human capital depreciation is the input, AI-related career anxiety is the error signal of a cybernetic control loop, and occupational adaptation or career avoidance is how the loop closes. Three properties follow — transmission, saturation, and conjunction — and each is tested on a two-wave survey of 1,247 Chinese graduates within three years of graduation (914 matched across waves, outcomes measured six months after their antecedents).

The highlights of the paper are as follows.

(1) The first inverted-U evidence on AI career anxiety. Anxiety converts into reskilling, credential pursuit and occupational repositioning up to a turning point at 0.37 standard deviations above the mean — established by the Lind–Mehlum composite test with a Fieller interval, not by the sign of a quadratic — and 38.5 percent of the cohort is already beyond it, in the region where more anxiety predicts less adaptation. Career avoidance, by contrast, rises monotonically. The same error signal that steers within a bounded band buys escape without bound.

(2) The conduit reverses sign, as a control-loop reading predicts. Bootstrapped conditional-process estimates show the identical depreciation percept, transmitted through anxiety, raising adaptation by 0.262 for a calm, supported, AI-literate graduate and lowering it by 0.281 for an anxious, unsupported, low-literacy one. Mediation whose sign depends on the state of the mediator is exactly what saturation of an error signal implies, and it has not previously been documented in this literature.

(3) The two levers sit where systems theory says they sit. Generative-AI literacy does not flatten the curve; it moves the turning point, from 0.07 to 0.73 standard deviations — capability extends the range over which the signal stays productive. Employability support acts a stage earlier, cutting the depreciation-to-anxiety pass-through from 0.64 to 0.33. Sensor recalibration and actuator range, estimated separately.

(4) Equifinality fails, informatively. Fuzzy-set qualitative comparative analysis returns exactly one sufficient recipe for high adaptation — anxiety AND literacy AND support jointly (consistency 0.892), every condition core, no consistent recipe for the negation, and no single condition necessary. Where configurational studies routinely report many roads, a tight control-loop theory predicted one road, and the data delivered it. Measurement quality is documented throughout (CFI 0.994, RMSEA 0.018; invariance across gender and major; three method-bias diagnostics), and the curvilinear result survives a behavioral hours outcome, attrition weighting, marker partialling and a wild-cluster bootstrap.

We believe the manuscript fits the Special Issue precisely. It answers the call’s question about feedback loops, emergent behaviour and adaptive responses to generative AI not by metaphor but by estimation: the loop’s stages, its saturation point, its boundary conditions and its conjunctural logic are each a numbered result. The practical payoff is a relocation of the policy lever. Two-fifths of a record graduating cohort already carries more career anxiety than it can convert into action; reassurance campaigns damp a signal that is in fact informative, while embedding generative-AI practice in curricula and building visible transition infrastructure complete the loop around it. We hope the manuscript proves suitable for the Special Issue, and we would be pleased to revise it in light of the reviewers’ comments.

Yours sincerely,

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